

Methods & Overloading: Car.java

Car.java (Part 1):

```
// Car.java
public class Car {
    public String make;
    public int year;
    // New variables to track mileage and oil changes
    private int milesDriven = 0;
    private int milesSinceOilChange = 0;

    public Car() {
        // Default constructor from previous assignment.
    }

    // Method to simulate the engine revving.
    public void revEngine() {
        System.out.println("VROOM!");
    }

    // Method to simulate driving the car.
    public int ride(int miles, int speed) {
        // Add miles driven to the counter.
        this.milesDriven += miles;
        this.milesSinceOilChange += miles;

        // Rev the engine before riding.
        revEngine();

        // Calculate and return the time in minutes.
        // Formula: time = (distance / speed) * 60 minutes
        // We use a double for accurate calculation before casting to int.
        double timeInHours = (double) miles / speed;
        return (int) (timeInHours * 60);
    }

    // Method to check if an oil change is needed.
    public boolean checkOil() {
        return this.milesSinceOilChange >= 5000;
    }

    // Method to reset the oil change counter.
```

```

        public void changeOil() {
            System.out.println("Oil change complete. Miles since last change
reset.");
            this.milesSinceOilChange = 0;
        }
    }
}

```

Car.java (Part 2):

```

// Car.java
public class Car {
    public String make;
    public int year;
    // Variables for method functionality
    private int milesDriven = 0;
    private int milesSinceOilChange = 0;

    // Overloaded constructors
    // Default constructor
    public Car() {
        this("Unknown", 0);
    }
    // Constructor with make and year
    public Car(String make, int year) {
        this.make = make;
        this.year = year;
    }
    // Constructor with make only
    public Car(String make) {
        this(make, 0);
    }
    // Constructor with year only
    public Car(int year) {
        this("Unknown", year);
    }

    // Method to simulate the engine revving.
    public void revEngine() {
        System.out.println("VROOM!");
    }

    // Overloaded ride method (with speed)
    public int ride(int miles, int speed) {

```

```
        this.milesDriven += miles;
        this.milesSinceOilChange += miles;
        revEngine();
        double timeInHours = (double) miles / speed;
        return (int) (timeInHours * 60);
    }

    // Overloaded ride method (without speed)
    public void ride(int miles) {
        this.milesDriven += miles;
        this.milesSinceOilChange += miles;
        revEngine();
        System.out.println("Drove " + miles + " miles. Time not calculated.");
    }

    public boolean checkOil() {
        return this.milesSinceOilChange >= 5000;
    }

    public void changeOil() {
        System.out.println("Oil change complete. Miles since last change
reset.");
        this.milesSinceOilChange = 0;
    }
}
```